

James Mann

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Education

University College London	Computer Science BSc	Sept 2024 – May 2027
• First year average: 81% • First Year Representative of the AI Society • President and founder of the AI Safety Society		
Carmel College	A Levels	Sept 2017 – May 2024
• A*A*A*AA (Maths, Further Maths, EPQ, Physics, Computer Science) • Elected Head Student		

Experience

Arcadia Impact	Research Analyst / Software Engineer	May 2025 -
• Led the creation of a technical tender responding to a €1,000,000 tender for AI Evaluations Infrastructure. • Part of the maintenance team for Inspect Evals, responding to issues and applying QA to prospective new Evaluations. • Independently developed cost modeling software that simulated 10,000+ instances of the company, accounting for underlying randomness, leading to 15% better resource allocation.		
Impact Research Groups	Technical AI Safety Research Sprint	Feb 2025 - April 2025
• First author of a paper produced as part of an 8 week competitive research sprint, coming third. • Identified a safety-critical latent factor in high-dimensional model activation space that could steer between refusal and compliance. • Proposed a theoretical framework for the success of jailbreaks as suppressors of harmfulness perception.		

Projects

LLM Foundational Research

- Utilised multivariate regression to identify and controlling planning capabilities in model latent space.
- Linear probes trained on input activations to predict the non-deterministic, sampling-induced distribution, utilizing the Kullback-Leibler divergence function, achieving a $0.95 R^2$.
- Applied in-place, input activation steering to control the high-level length feature i.e. make the model output longer responses, or make the model output shorter responses, without affecting coherence.

GPT from Scratch

- Created a transformer from scratch using NumPy with modular attention and multi-layer perceptron blocks.
- With no additional dependencies designed an implementation of gradient descent and backpropagation to optimise the >100,000,000 parameters of the model for next token prediction.

Probabilistic Image Generation

- Implemented a denoising diffusion probabilistic model in Pytorch, trained on CIFAR100 to generate novel images from the distribution, iteratively optimising the hyperparameters to improve the plausibility of samples.
- Followed the paper Denoising Diffusion Probabilistic Models, implementing a UNet with attention heads

Interests

Rowing: Compete for the university in the novice A boat.

French: Self teaching the language with an interest to study abroad.